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COUNTY-LEVEL RISK STRATIFICATION FOR MANAGED CARE OPTIMIZATION

OBJECTIVE

Design a geographic risk segmentation model integrating chronic disease burden and social determinants of health (SDOH) to identify structural care risk patterns across 2,957 U.S. counties using CDC PLACES data.

METHODOLOGY

- Aggregated chronic disease prevalence measures (diabetes, COPD, obesity, hypertension, etc.) into a composite clinical burden index.
- Aggregated SDOH-related indicators (food insecurity, housing instability, transportation access, social support) into a composite social risk index.
- Standardized both domains to percentile-based 0–100 risk scores.
- Constructed a weighted composite risk model (60% chronic, 40% SDOH).
- Segmented counties into High / Medium / Low risk tiers.
- Quantified structural divergence between clinical burden and social vulnerability.

KEY FINDINGS

- 29.9% of counties fall within the High composite risk tier.
- Chronic burden and SDOH risk demonstrate moderate positive correlation ($r \approx 0.50$).
- Mean divergence between domains is 22.6 percentile points, indicating meaningful structural imbalance across many regions.
- 28% of counties exhibit severe (>30-point) chronic–social misalignment, revealing concentrated intervention opportunities.

Risk + Chronic Burden
≈ **0.5**
Moderate positive correlation (r = 0.50)

Structual Imbalance
22.6
point mean divergence

Counties at high-risk
29.9%
of counties

Counties in need of targeted interventions
28%
of counties

OPERATIONAL RELEVANCE

- Prioritize geographic deployment of care management resources.
- Identify regions where social vulnerability outpaces clinical burden.
- Detect structural misalignment associated with preventable utilization risk.
- Inform value-based contracting and network design strategy.

KEY INSIGHT

Risk is not randomly distributed — it clusters geographically and diverges structurally, revealing measurable opportunities for targeted managed care intervention.